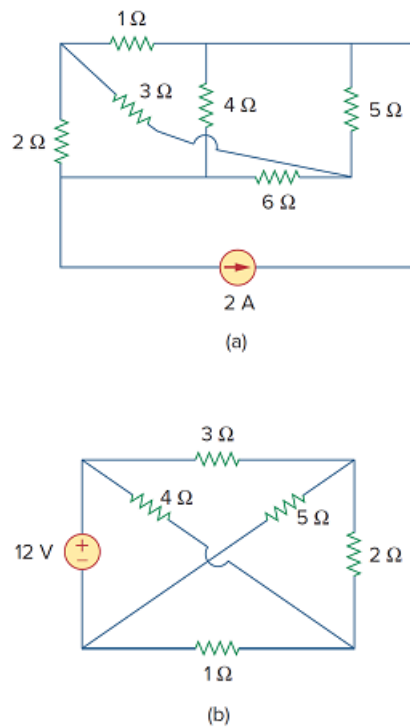


Problem

Which of the circuits in Fig. 3.82 is planar? For the planar circuit, redraw the circuits with no crossing branches.

Figure 3.82:



Step-by-step solution

Step 1 of 4

(a)

Refer to the circuit diagram in Figure 3.82 in the textbook.

Redraw the circuit for node identification.

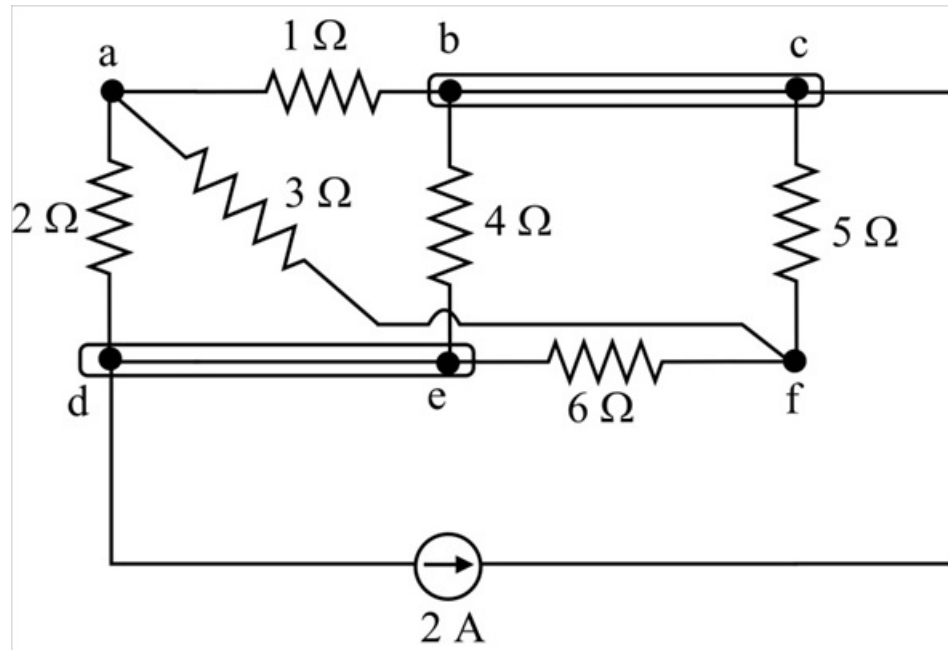


Figure 1

Step 2 of 4

Try to draw figure 1 such that it has no crossing branches.

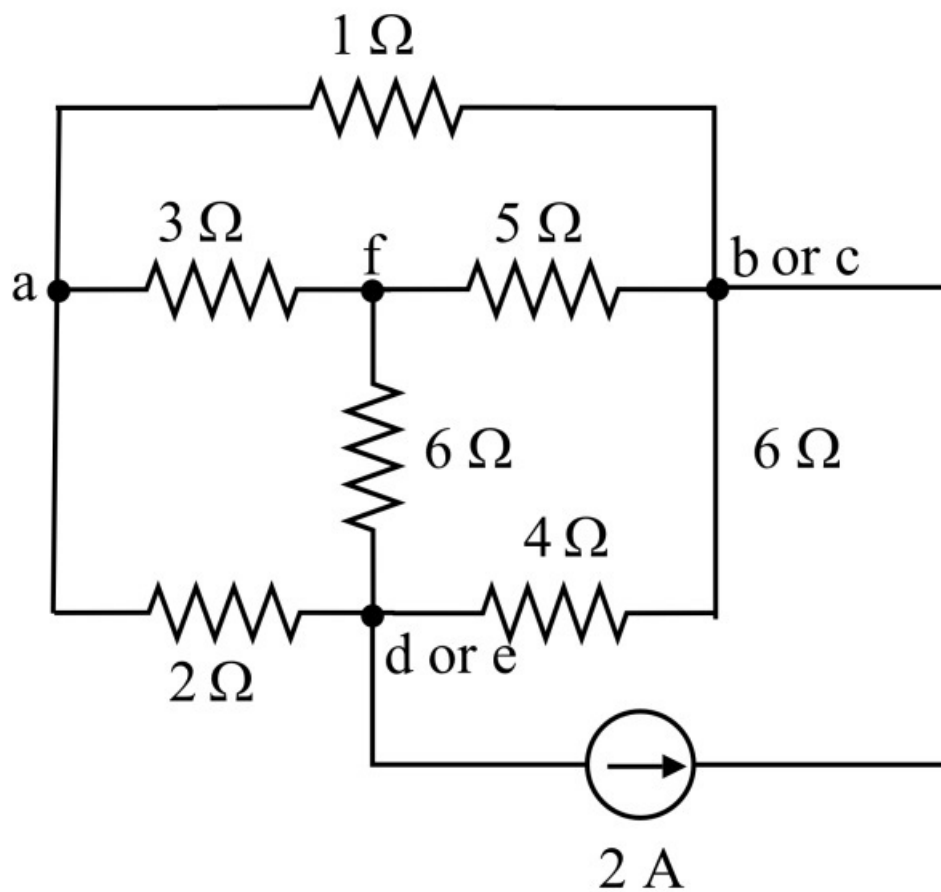


Figure 2

From Figure 2, it is possible to draw Figure 1 without crossing branches. So the circuit is **planar**.

(b)

Refer to the circuit diagram in Figure 3.82(b) in the textbook.

Redraw the circuit for node identification.

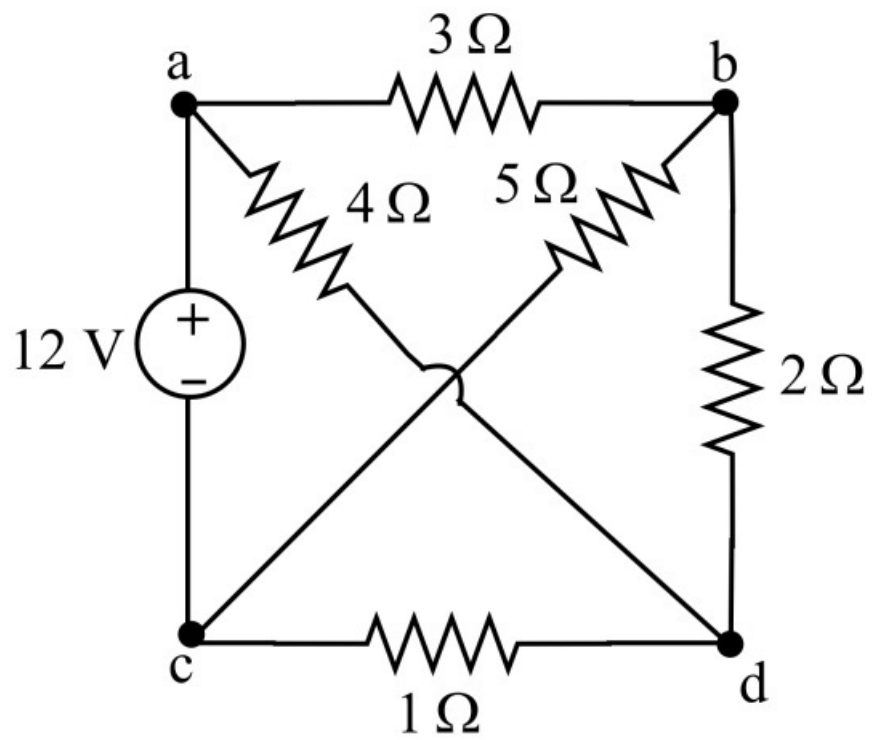


Figure 3

Try to draw figure 3 such that it has no crossing branches.

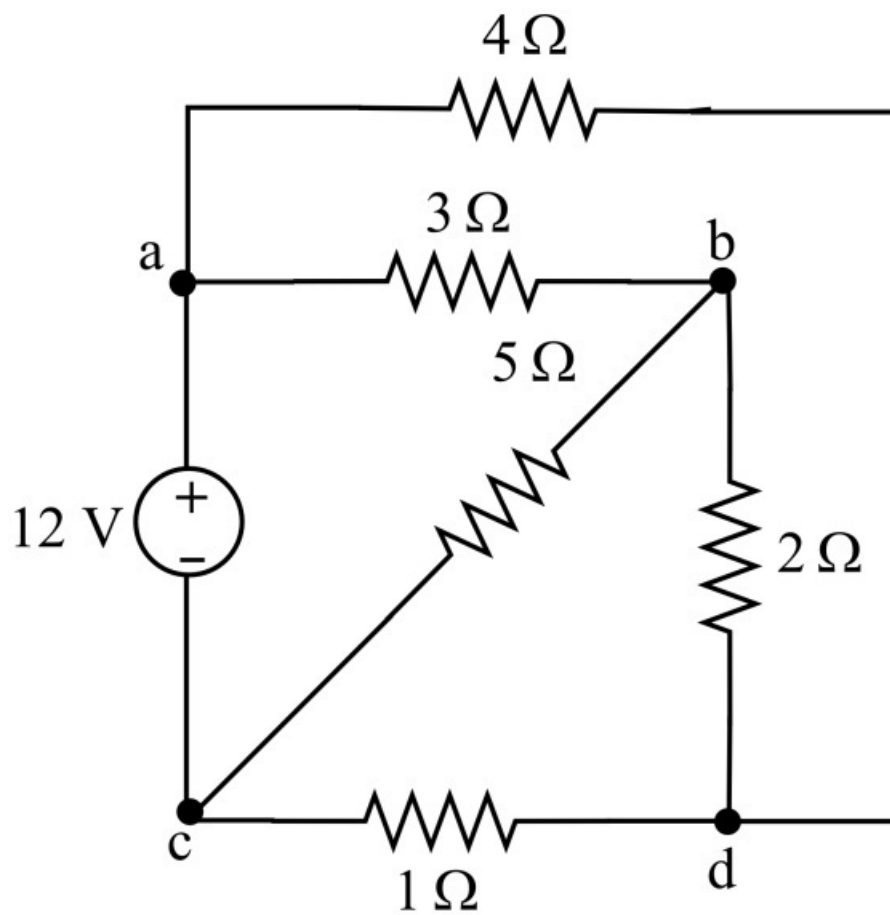


Figure 4

From Figure 4, it is possible to draw Figure 3 without crossing branches. So the circuit is **planar**.